LIAM CAWLEY

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EDUCATION

UNIVERSITY OF MICHIGAN, **College of Engineering** May 2025
Bachelor of Engineering, Computer Science Engineering, Minor in Mathematics

COURSEWORK

• Random Numerical Linear Algebra • Principles of Machine Learning • Science of Language Models • Game Theory • Linear Algebra • Graph Algorithms • Mathematics of Finance • Probability Theory • Randomness and Computation • Matrix Algebra • Computer Vision

EXPERIENCE

Machine Learning Engineer May 2025 – Present
KLA Corporation *Ann Arbor, MI*

- Led backend implementation of a production ETL and inference pipeline (DAL) that served proprietary encrypted vision models as a child of a legacy 12-microservice, 8 compute node Docker swarm (DART) deployed to on prem 12-GPU rack.
- Developed a hybrid local/remote inter-process communication model, utilizing POSIX Shared Memory (/dev/shm) for zero-copy local fast-path IPC on single-node GPU workers, and gRPC for cross-node inter-service communication.
- Led release engineering for target-node deployments, including appsettings configuration, dependency validation, service health checks, and CI/CD orchestration across Docker Swarm nodes.
- Built an AI analysis service with Gemma-4-26B-A4B served via vLLM container with TensorRT (orchestrated with LangGraph) with multiple subagents, RAG, in-memory session management, and 20 integrated tools with realtime DART service integration.
- Awarded 1st Place at KLA's global algorithms conference for developing a Physics-Informed Neural Network system and presenting its technical roadmap to senior division leadership.

High Performance Computing Intern May 2024 – Aug 2024
KLA Corporation *San Jose, CA*

- Built a proprietary vision model system as a proof of the hypothesis that physical chip layout geometries could be associated with systematic occurrences of atomic-scale transistor defects.
- Co-authored a high-performance design-geometry parser by extending KLA's internal **GDSTK** C++ engine, optimizing the extraction and localization of layout structures for specific wafer defect locations.
- Implemented custom CUDA rasterization and defect-centered cropping operations bound to Python via **pybind11**, accelerating geometry-to-tensor rendering on the GPU and bypassing slow CPU-to-GPU memory copies.
- Engineered an asynchronous data loader using **NVIDIA DALI** to pipeline and overlap file prefetching and GPU tensor transfers with active deep learning inference via concurrent CUDA streams.
- Demonstrated that design-derived feature vectors performed on-par with legacy physical optical metrics, significantly improving the classification accuracy of production Gradient Boosted Decision Tree (GBDT) ensembles.

Machine Learning Intern May 2023 – Aug 2023
EMAG Technologies, Inc. *Ann Arbor, MI*

- Built a closed-loop calibration application for active electronically scanned phased-array radar, optimizing 6-bit discrete phase-shifter and attenuator states on Anokiwave AWS-0103 beamforming ICs.
- Engineered the optimization loop around physical latency constraints, utilizing metaheuristic search algorithms (Genetic Algorithms / PSO) integrated with a software-state caching layer to prevent redundant SPI programming and SDR digitizing runs.
- Developed a CNN-based physical surrogate model that treated the 16×16 aperture array as a 2D grid, learning mutual coupling and non-linear AM/PM distortion to predict optimal phase adjustments in software and bypass slow physical measurements.
- Achieved a **6.5× reduction** in post-calibration beam synthesis error compared to the baseline Sample Matrix Inversion (SMI) method under representative hardware-in-the-loop test configurations.

PROJECTS

Autonomous Fixed-Wing UAV with Object Classification C++, Python, OpenCV, ArduPilot, QGroundControl

- Built a fully autonomous 1.6m fixed-wing UAV (MFD Crosswind Mini) for the AUVSI search-and-rescue competition; integrated Pixhawk Cube Orange, RTK GPS, dual-camera payload, and 900 MHz telemetry into a CG-constrained airframe.
- Tuned ArduPilot's **EKF3 state estimation** and closed-loop control loops (**PID / TECS**) to achieve stable autonomous takeoff, waypoint-navigation, and failsafe maneuvers under variable wind conditions.
- Developed an Object Detection, Classification, and Localization (ODCL) pipeline utilizing a **YOLOv5** model to detect and geolocate targets in real-time from high-resolution Sony A6000 aerial imagery.

SELECTED OPEN-SOURCE / RESEARCH

MetaRep (ICL ↔ Kernel Ridge): theory + experiments validating ridge-like ICL; reproducible harnesses and proofs.
StableGLM (Rashomon sets for GLMs): ε-Rashomon membership/certificates and set-level interpretability metrics.

SKILLS

Programming Languages: Python, C++, C, C, TypeScript, Java, SQL, MATLAB, LaTeX, Mojo
HPC & Acceleration: CUDA, NVIDIA DALI, pybind11, Nsight Systems, gRPC / Protocol Buffers, GDSTK, OpenMP
ML & Infrastructure: PyTorch, TensorFlow, Docker, Kubernetes, Docker Swarm, Postgres, Git, XGBoost / GBDTs
Activities: UM HuggingFace Admin, UM Hackers Quant Team, FPV Building/Racing, Archery